Reg No.

INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-1

Programme: B.E. ECE-DS | Max. Marks: 50
Course Code: SMTB1213 | Time: 2 Hours
Course Name: Discrete Structures and Numerical Methods | Sem: II
Batch : 2025-2029 | Date: 19.02.2026

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Define tautology	1
2.	Verify $P \wedge Q \Rightarrow Q$	1
3.	Verify $P \rightarrow Q \Leftrightarrow 7PVQ$	1
4.	Symbolize the statement "All men are mortal"	1
5.	Define transitive relation with an example	2
6.	Examine whether $R=\{(i, j) : \lvert i - j \rvert = 2\}$ on the set $A = \{1,2,3,4,5,6\}$ is symmetric	2
7.	Let $X = \{1, 2, 3, 4\}$ and let $R = \{(x, y) : x \leq y\}$. Construct the graph of R.	2

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Identify the PDNF and PCNF of $(P \wedge Q) V (7P \wedge R) V (Q \wedge R)$	1
	(OR)	
9.	Verify that the set of premises $P \rightarrow Q$, $P \rightarrow R$, $Q \rightarrow 7R$ and P is inconsistent.	1
10.	Show that $(x)(P(x) \rightarrow Q(x)) \wedge (x)(Q(x) \rightarrow R(x)) \Rightarrow (x)(P(x) \rightarrow R(x))$	1
	(OR)	
11.	Let $X = \{2, 3, 4, 6, 12, 24, 36\}$ and the relation R be defined as xRy iff "x divides y". Construct the Hasse Diagram for the partial ordering relation R on X.	2
12.	Verify that the relation "congruence modulo m" over the set of positive integers is an equivalence relation	2
	(OR)	

13.	If R and S are any two relations on a set A, verify (i) whenever R and S are reflexive, R∩S and RUS are also reflexive (ii) whenever R and S are transitive, R∩S is also transitive but RUS need not be transitive	2